RAKESH CHOWDARY POLINA

rpolina@okstate.edu | https://www.linkedin.com/in/rakesh-chowdary-polina | 405-894-3989

EDUCATION

Master of Science in Computer Science **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

SKILLS

Languages:	Python, Java, JavaScript, SQL, C, C++, HTML, CSS, Swift
Frameworks:	Flask, Django, React JS, Angular JS, Node JS, Spring boot, FastAPI, React Native
Databases:	MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra
Web Technologies:	RESTful API, Django ORM, SQLAlchemy, OAuth 2, jQuery, JMS, RabbitMQ
Web Services:	REST, SOAP
Tools/Platforms:	Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger
AWS Cloud Services:	EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront
Build Tools:	Pip, Poetry

WORK EXPERIENCE

Research Data Analyst, Oklahoma State University **May 2023 – May 2024**

- Engineered a facial recognition pipeline using Python and InsightFace, performing quantitative analysis on 182,000+ institutional images to achieve 93.6% precision. Integrated Ellucian Banner ERP data (student/faculty records) to validate identity metadata, ensuring alignment with institutional databases.
- Improved data reliability by implementing DBSCAN clustering to detect and resolve anomalies in image archives, enhancing research accuracy by 22%. Automated workflows using SQL and Python to reconcile discrepancies with Ellucian ERP datasets.
- Developed a Flask-based internal tool with React.js for streamlined image labeling and annotation, reducing manual effort by 35%. Configured JSON/XML outputs for compatibility with downstream systems, including Ellucian-powered reporting dashboards.
- Designed and maintained SQL Server databases to manage user interactions and feedback, ensuring data integrity for research projects. Optimized cross-system queries between research databases and Ellucian ERP modules (e.g., admissions, HR) using PL/SQL procedures.
- Collaborated with institutional IT to troubleshoot integration issues between analytics tools and the Ellucian environment, resolving 15+ API/data synchronization errors through log analysis and Bash scripting.

Software Engineer, Ericsson, India **Jul 2019 – Jul 2022**

- Transitioned monolithic systems to microservices using Python, Flask, and FastAPI, designing RESTful APIs with JSON/XML output formats. Integrated vendor-provided APIs for third-party services and implemented asynchronous communication with Apache Kafka, improving scalability to support 50% more concurrent users. Optimized relational database performance by developing PL/SQL stored procedures and refining complex SQL queries.
- Developed full-stack web solutions using JavaScript (React) and PHP, enhancing user interfaces for internal dashboards and customer portals. Implemented dynamic content rendering and responsive designs with TypeScript, reducing front-end load time by 25%.
- Engineered Python-based backend systems with Django, optimizing PostgreSQL data access layers. Reduced system downtime by 30% through query optimization, indexing, and security enhancements like role-based access control (RBAC) and data encryption.
- Designed and maintained RESTful APIs compliant with enterprise standards, achieving 40% faster response times using asynchronous programming. Integrated real-time data updates using WebSocket and formatted responses in JSON/XML for cross-platform compatibility.
- Automated deployment pipelines using Bash/Shell scripting with Jenkins and GitLab CI/CD, reducing deployment time by 40%. Incorporated AWS CodeBuild for pipeline efficiency and wrote scripts for automated database migrations and log analysis.
- Troubleshooted and resolved production issues by analyzing SQL query execution plans, application logs, and API metrics. Collaborated with DevOps to streamline monitoring using Grafana, reducing incident resolution time by 35%.
- Enforced security best practices, including OAuth 2.0 authentication and TLS encryption, ensuring compliance with AWS and organizational standards.
- Followed Agile (Scrum) methodologies with bi-weekly sprints, daily standups, and Jira-based task tracking. Collaborated with cross-functional teams to align API specifications with business requirements.
- Authored technical documentation for APIs, database schemas, and deployment processes, ensuring clarity for stakeholders and seamless knowledge transfer.

PROJECTS

Sleep data analysis - Applied LSTM network and Granger causality techniques to investigate sleep data on a dataset comprising over 5,000 individuals' sleep patterns, unveiling insights into temporal dynamics and causal connections within the dataset